It’s Not Easy Being Wheezy! Predicting Asthma Risk from Hidden Pollutants

Inspiration

Air pollution is a well documented health risk, with especially strong impacts on the 28 million Americans living with asthma. Many people rely on the Air Quality Index (AQI) to understand daily risk, but standard AQI only reflects the worst pollutant and misses combined multi-pollutant exposure effects. Understanding which pollutants have the strongest impact on emergency room visits and whether this impact varies by age, requires rigorous data analysis. Here, we move beyond correlation and build predictive models which quantify these relationships. Using our interactive GUI, California users can better understand county-level asthma risk, enabling more informed public health decisions.

What it does

We analyzed EPA air quality data alongside California's asthma emergency department (ED) visit data from 2015 to 2022. While AQI summarizes air quality using a single worst-pollutant index, our approach models individual pollutants simultaneously to capture multi-exposure effects. We created predictive models that identify which pollutants are most strongly associated with asthma ED visit rates and used multivariate models to evaluate how these effects vary between children and adults across California.

How we built It

1. **Exploratory Data Analysis:**

   Started with 16 air quality parameters but discovered limited sample sizes. Focus our analysis on pollutants with sufficient observations.

2. **Single Pollutant Models:**

   Ran univariate linear regressors for each pollutant, retaining those with a p-value < 0.05 and adequate sample sizes.

3. Multivariate Linear Regression:

   Built multivariate linear regression model using ozone, nitric oxide (NO), and nitrogen dioxide (NO_2), as primary predictors, with age group as a moderator. Compared raw

versus within-county demeaned inputs to distinguish absolute pollution exposure effects from local variations. Model explains 53% of variance in asthma-related ED visits (R^2 = 0.532).

4. **Validation of Multivariate Regression**:

   5-fold cross-validation of our model performance (Mean R^2 = 0.512 ± 0.053 across 5 folds) revealed robust generalization with minimal overfitting (training R^2 0.532 vs. CV R^2 0.512).

5. **Multivariate Machine Learning (XGBoost):**

   Trained an XGBoost model to capture nonlinear interactions among key pollutants. Evaluated performance with R^2 (0.3149), feature importance, and SHAP (SHapley Additive exPlanations). While XGBoost captured nonlinear structure, the multivariate regression model ultimately provided comparable performance and was selected as the final model due to its interpretability.

Main Analysis Results

Multivariate Regression

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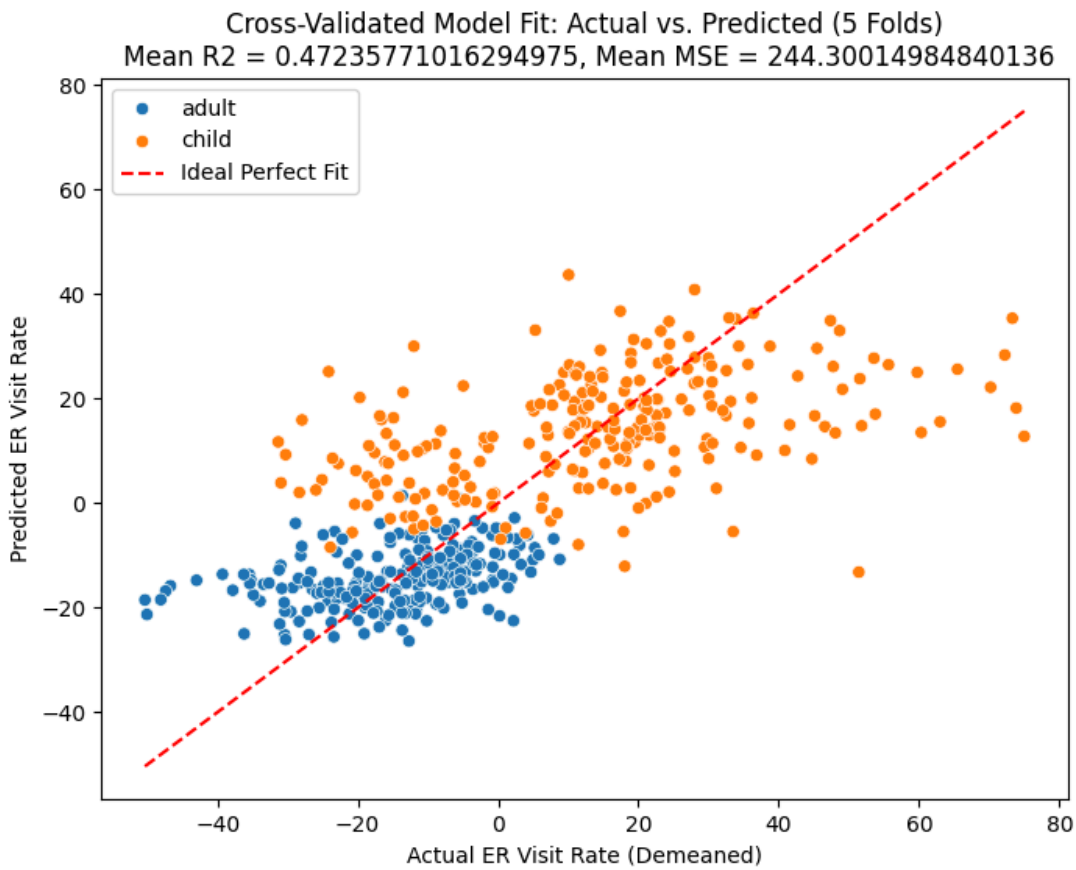
After data imputation, still have to drop 262 entries due to one or moremissing pollutant data.
                                OLS Regression Results
==============================================================================
Dep. Variable:     AGE_ADJUSTED_ED_VISIT_RATE   R-squared:                       0.532
Model:                                    OLS   Adj. R-squared:                  0.526
Method:                         Least Squares   F-statistic:                     79.04
Date:                        Sun, 19 Apr 2026   Prob (F-statistic):           3.57e-76
Time:                                10:15:00   Log-Likelihood:                -2049.1
No. Observations:                         494   AIC:                             4114.
Df Residuals:                             486   BIC:                             4148.
Df Model:                                   7
Covariance Type:                    nonrobust
==========================================================================================================
                                              coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------------------------------
Intercept                                -14.0603      0.983    -14.307      0.000     -15.991     -12.129
C(AGE)[T.child]                           28.1206      1.390     20.234      0.000      25.390      30.851
Q("Ozone")                              1958.9908    867.281      2.259      0.024     254.907    3663.075
Q("Ozone"):C(AGE)[T.child]              3469.6969   1226.521      2.829      0.005    1059.758    5879.636
Q("Nitric oxide (NO)")                     2.4234      1.398      1.733      0.084      -0.324       5.171
Q("Nitric oxide (NO)"):C(AGE)[T.child]     3.5396      1.978      1.790      0.074      -0.346       7.425
Q("Nitrogen dioxide (NO2)")                3.2750      2.186      1.498      0.135      -1.021       7.571
Q("Nitrogen dioxide (NO2)"):C(AGE)[T.child]  0.9727      3.092      0.315      0.753      -5.103       7.048
==============================================================================
Omnibus:                       24.278   Durbin-Watson:                   1.692
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               54.462
Skew:                           0.234   Prob(JB):                     1.49e-12
Kurtosis:                       4.558   Cond. No.                     3.14e+03
==============================================================================
```



Metrics for each fold:

This fold: R^2 = 0.536, MSE = 230.537

This fold: R^2 = 0.555, MSE = 240.347

This fold: R^2 = 0.409, MSE = 314.106

This fold: R^2 = 0.523, MSE = 237.083

This fold: R^2 = 0.537, MSE = 200.539

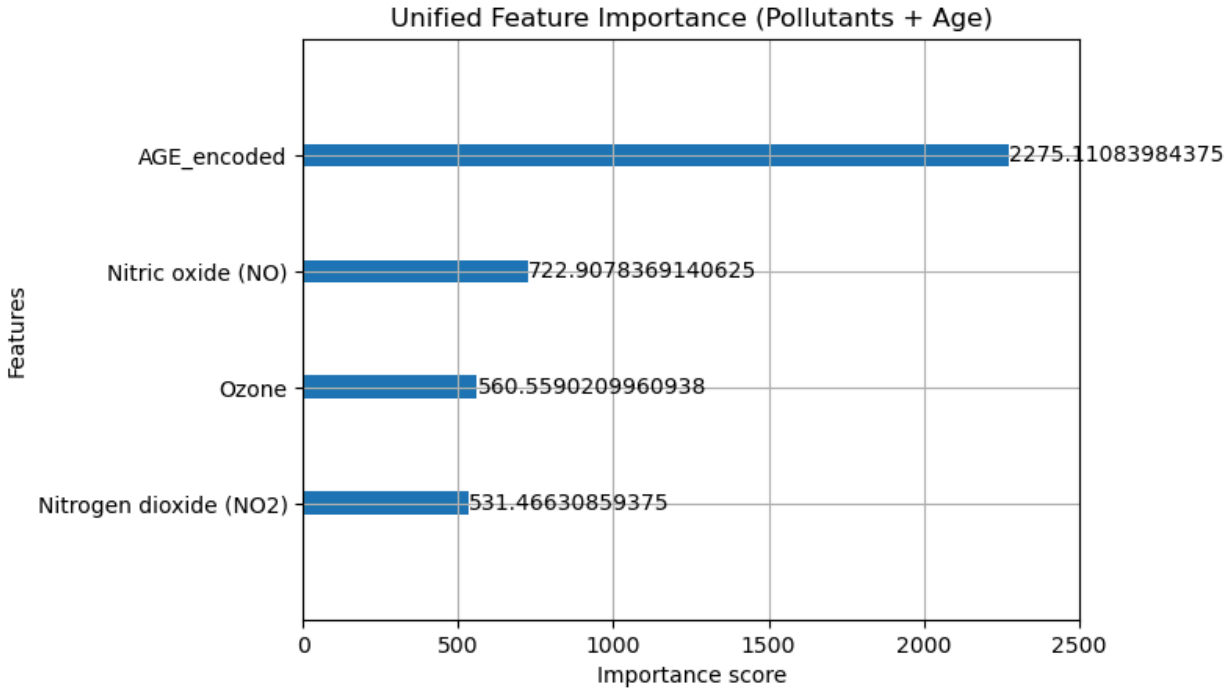
========== Final CV Performance ==========

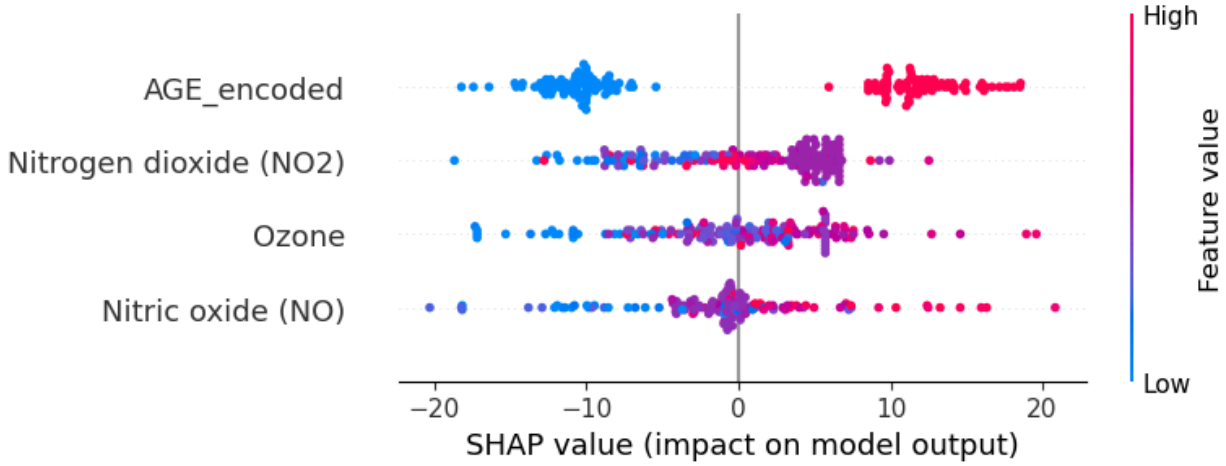
Mean R^2: 0.512 ± 0.053

Mean MSE: 244.522 ± 37.536

XGBoost & SHAP

Unified Model R^2: 0.3149
Unified Model RMSE: 20.4319





Challenges we ran into

A major challenge was aligning datasets across different sources, timescales, and reporting structures. Across all datasets, there was substantial missing data, which impacted the power and direction of our statistical analyses.

Accomplishments that we're proud of

We successfully developed a multivariate regression model that captures multi-pollutant exposure and cross-validated our model performance. We successfully translated our findings into an intuitive dashboard designed to make complex air quality and its health impacts accessible to individuals with asthma.

What we learned

We learned about AQI and how different pollutants are measured by the EPA. We also gained experience working with large public health datasets, running different types of regression, and creating a user-friendly visualization tool.

What's next for It’s Not Easy Being Wheezy!

In the short term, we will investigate collinearity between nitrogen oxide species (NO, NO_2, NOy) to improve the model stability. Beyond this, we aim to extend our work into a real-time risk prediction tool by integrating with at-home air monitoring systems, like Purple Air.

APA References

California Health and Human Services Agency. (n.d.). *Asthma emergency department visit rates* [Dataset]. Retrieved from https://data.chhs.ca.gov/dataset/asthma-emergency-department-visit-rates

U.S. Environmental Protection Agency. (n.d.). *Air quality system (AQS): Annual AQI quality data* [Dataset]. Retrieved from https://aqs.epa.gov/aqsweb/airdata/download_files.html